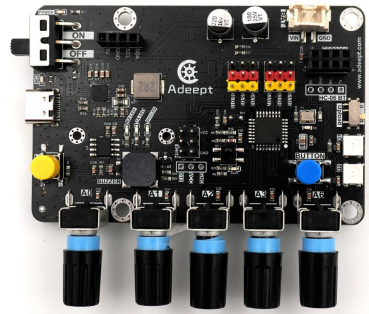


Lesson 8 How to Use the Infrared Module

8.1 Overview

This course mainly introduces the relevant knowledge of infrared modules, including their working principles, hardware connections, and programming applications on the Arduino platform. Be able to master the working mechanism of the transmitting and receiving parts in infrared remote control systems, learn to use infrared modules to achieve remote control functions of equipment, and lay the foundation for developing various infrared control projects.

8.2 Required Components

Components	Quantity	Picture
Adeept Robotic Arm Control Board	1	
Type-C USB Cable	1	
IR remote	1	
OLED Screen	1	

8.3 Principle Introduction

Infrared control is generally divided into two parts, namely the transmitting part and the receiving part, the IR receiver belongs to the receiving part, and the IR remote control belongs to the transmitting part.

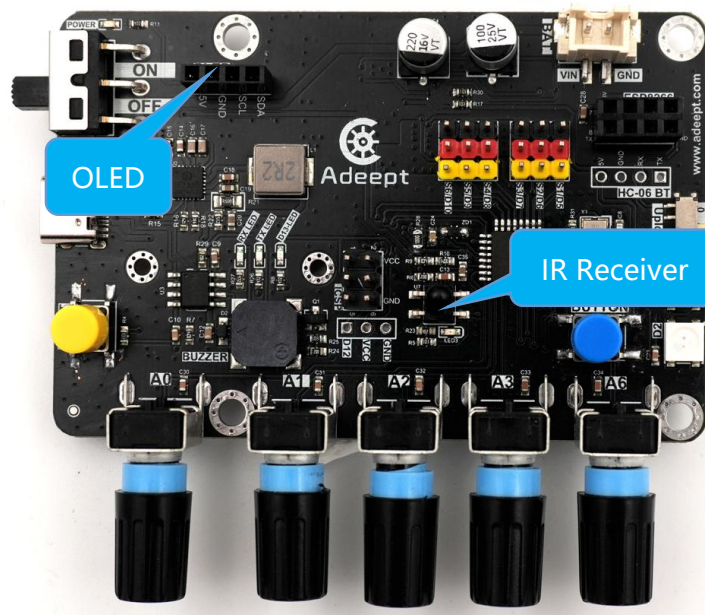
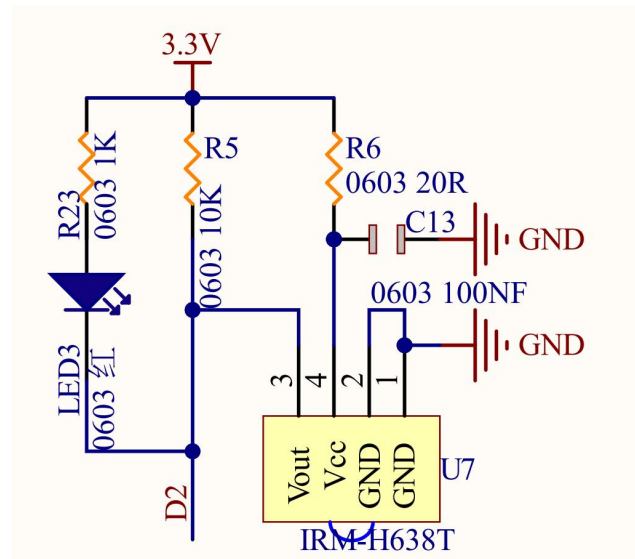
There are two important components inside the IR receiver, namely IC and PD. IC is the processing element of the receiving head. It is mainly composed of silicon crystals and circuits. It is a highly integrated device. Its main functions include filtering, shaping, decoding, and amplification. PD is a photodiode, and its main function is to receive optical signals.

The transmitting circuit of the IR remote control uses infrared light-emitting diodes to emit modulated infrared light waves. Press a certain key of the remote control, the remote control will send out a series of modulated signals. After the signal is received by the infrared integrated module, it outputs the demodulated digital pulse. Each key corresponds to a different pulse, so it can identify different pulses. pulses to identify different keys.

The following is the information in the corresponding program when each button of the infrared remote control is pressed.



8.4 Wiring Diagram



PINS of Adept Robotic Arm Control Board	IR Receiver
D2	VOUT
3.3V	VCC

GND	GND
GND	GND

8.5 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board (Arduino Board) with a USB cable.
2. Open "**08_IR_Receiver**" folder in **"/Code"** , double-click "**08_IR_Receiver.ino**" .



```

08_IR_Receiver | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
08_IR_Receiver.ino
1  /*****
2   Filename   : 08_IR_Receiver.ino
3   Author    : www.adeept.com
4   *****/
5   #include <Wire.h>
6   #include <SSD1306AsciiWire.h>
7   #include <IRremote.h>
8   SSD1306AsciiWire display;
9   #define RECV_PIN 2
10  String command_ir = "";
11
12  void setup(){
13    // Initialize OLED
14    Wire.begin();
15    Wire.setClock(400000L);
16    display.begin(&Adafruit128x64, 0x3C);
17    display.setFont(Adafruit5x7);
18    display.clear();
19    display.set2X();
20    display.setCursor(0, 0);
21    display.print("Receive:");

```

3. Select development board and serial port.

Board: **Tools---**>**Board---**>**Arduino AVR Boards---**>**Arduino Uno**

Port: **Tools --->Port---**>**COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```

Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.

```

5. After successfully uploading the code, run the program. Use the IR remote to aim at the IR receiver on the expansion board. Press a button, and the corresponding button information can be displayed on the screen.

```

Output  Serial Monitor  X
Message (Enter to send message to 'Arduino Uno' on 'COM3')  New Line  115200 baud
IRrecvDemo is now running and waiting for IR message on Pin 2
A
B
C

```



8.6 Code

```

01  /*****
02  Filename   : 08_IR_Receiver.ino
03  Author    : www.adeept.com
04  *****/
05  #include <Wire.h>
06  #include <SSD1306AsciiWire.h>
07  #include <IRremote.hpp>
08  SSD1306AsciiWire display;
09  #define RECV_PIN 2
10  String command_ir = "";
11
12  void setup(){
13    // Initialize OLED
14    Wire.begin();
15    Wire.setClock(400000L);
16    display.begin(&Adafruit128x64, 0x3C);
17    display.setFont(Adafruit5x7);
18    display.clear();
19    display.set2X();
20    display.setCursor(0, 0);
21    display.print("Receive:");

```

```
22
23   Serial.begin(115200);
24   IrReceiver.begin(RECV_PIN, ENABLE_LED_FEEDBACK);
25   Serial.print("IRrecvDemo is now running and waiting for IR message on Pin ");
26   Serial.println(RECV_PIN);
27 }
28
29 void loop(){
30   if (IrReceiver.decode()) {
31     unsigned long value = IrReceiver.decodedIRData.command;
32     command_ir = switch_irr(value);
33     Serial.println(command_ir);
34
35     display.setCursor(0, 3);
36     display.print("      ");
37     display.setCursor(0, 3);
38     display.print(command_ir);
39     IrReceiver.resume();
40   }
41   delay(100);
42 }
43
44 String switch_irr(unsigned long irr_data){
45   switch(irr_data){
46     case 69: return "A";
47     case 70: return "B";
48     case 71: return "C";
49     case 68: return "D";
50     case 64: return "UP";
51     case 67: return "E";
52     case 7: return "LEFT";
53     case 21: return "OK";
54     case 9: return "RIGHT";
55     case 22: return "0";
56     case 25: return "DOWN";
57     case 13: return "F";
58     case 12: return "1";
59     case 24: return "2";
60     case 94: return "3";
61     case 8: return "4";
62     case 28: return "5";
63     case 90: return "6";
64     case 66: return "7";
65     case 82: return "8";
66     case 74: return "9";
67     default: return "NONE";
68   }
69 }
```